



Original Contribution

Deep Learning Algorithm Based on Contrast-Enhanced Ultrasound Potentially Optimizes Treatment Strategies for Solitary Primary Hepatocellular Carcinoma

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ARTICLE INFO

Keywords:

Deep learning
Contrast-enhanced ultrasound
Hepatocellular carcinoma
Recurrence-free survival
Surgical resection
Thermal ablation

ABSTRACT

Objective: We aimed to propose a prognostic framework using a dual-branch Vision Transformer (ViT) deep learning (DL) architecture for stratifying recurrence risk in solitary primary hepatocellular carcinoma (HCC) patients undergoing surgical resection (SR) or thermal ablation (TA) and explore its potential value as a tool for aiding therapy-related discussions.

Methods: This two-center study enrolled 575 HCC patients with preoperative contrast-enhanced ultrasound (CEUS) images between January 2017 and October 2022. Among them, 365 underwent SR and 210 underwent TA. We combined ViT-extracted CEUS features with clinical features to separately establish models (ViT-SR and ViT-TA) for recurrence-free survival (RFS) stratification across SR and TA cohorts. Finally, Risk reclassification was further performed by applying the baseline-balanced patients to the alternative treatment model to investigate the potential for treatment optimization.

Results: The models exhibited favorable performance, achieving C-Index, AUC, precision, and F1 score of 0.73, 0.76, 0.86, and 0.70 in validation cohorts for ViT-SR, respectively, while corresponding values for ViT-TA were 0.76, 0.83, 0.76, and 0.76. Time-dependent AUC values demonstrated consistent stability across 1-/3-/5-y (ViT-SR: 0.76, 0.77, 0.76; ViT-TA: 0.79, 0.84, 0.82), with both models showing satisfactory calibration in the validation cohort. Treatment strategy conversion analysis indicated that 11.6% (17/146) of SR patients and 4.8% (7/146) of TA patients had a potentially more favorable prognostic profile under the alternative treatment.

Conclusion: ViT-based DL models that integrated multi-phase CEUS images and clinical variables facilitated non-invasive prediction of HCC recurrence probability, and may aid discussions on personalized clinical treatment strategies by simulating treatment conversion.

Introduction

Hepatocellular carcinoma (HCC) ranks among the most prevalent and lethal cancers globally, with substantial disease burden and mortality rates [1,2]. Consequently, a pivotal aspect of HCC management lies in achieving optimal therapeutic efficacy for early-stage tumors, which is crucial for improving patient outcomes. International guidelines classify solitary HCC ≤ 5 cm as either very early or early stage, for which surgical resection (SR) group and thermal ablation (TA) are considered standard treatments [2,3]. However, despite the widespread adoption of these two methods in clinical

practice, inconsistencies persist across various studies regarding their postoperative recurrence rates and long-term survival outcomes. A nationwide cohort study by Charalel et al. [4] found higher cancer-specific mortality and recurrence rates in ablation cohorts compared to surgical groups. In contrast, a single-center retrospective analysis demonstrated comparable overall survival (OS) rates between microwave ablation (MWA) and laparoscopic hepatectomy for solitary HCCs measuring 3–5 cm [5]. These discrepancies highlight the complexity of treatment decisions for early-stage HCC, with variations in efficacy potentially stemming from differences in study populations, tumor selection criteria, and specific technical

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implementation. Consequently, there is an urgent need for tools enabling more precise patient prognosis stratification to enhance treatment outcomes for HCC.

SR is generally considered the preferred treatment option for patients without significant portal hypertension and with preserved liver function. Its key advantage lies in enabling complete resection of the tumor-containing liver segment along with surrounding at-risk tissue, thereby achieving sufficient safety margins to reduce the risk of local recurrence. However, this approach carries a higher risk of postoperative complications due to its greater invasiveness. Furthermore, due to compromised hepatic reserve or unfavorable tumor location, only 9%–27% of HCC patients are suitable candidates for surgical treatment [6]. In contrast, TA may offer comparable survival outcomes, particularly for HCCs < 2 lt; 2 cm. It is especially suitable for patients with poor surgical tolerance, anticipated insufficient residual liver volume, or impaired liver function, while being more cost-effective and less invasive [7]. However, risks include interference from pulmonary gas during puncture and thermal injury to adjacent organs. Thus, selecting between SR and TA in clinical practice involves a delicate balancing act among these factors, making the decision challenging. Moreover, treatment options often overlap, meaning that when two choices are both technically feasible, making a decision can also be difficult. Therefore, we aimed to extract additional prognostic information from the tumor's own imaging characteristics and relevant clinical indicators to provide multidimensional support for personalized decision-making in HCC patients [3].

Previous studies demonstrated that the radiomic features of computed tomography (CT) and magnetic resonance imaging (MRI) have predictive value for prognostic stratification [8–10]. The potential utility of contrast-enhanced ultrasound (CEUS) in this regard remains to be fully investigated. Compared to CT or MRI, CEUS demonstrates superior temporal resolution, enabling real-time visualization of microcirculatory perfusion dynamics in HCC [11]. Additionally, traditional deep learning (DL) models exhibit inherent limitations in capturing global dependencies and long-range contextual relationships within images due to their localized convolutional operations [12]. The Vision Transformer (ViT) model, based on the transformer architecture, represents a paradigm shift in medical image analysis. Through its self-attention mechanism, ViT enables global feature interaction, effective fusion and effectively understands multi-source information, thereby providing more accurate information for prognostic assessment [13,14]. To the best of our knowledge, there is currently a lack of research investigating the application of Transformer models combined with CEUS for optimizing treatment strategies in early-stage HCC.

Therefore, this study aimed to develop and validate two ViT-based fusion models that integrate multi-modal ultrasound imaging and clinical variables for predicting the recurrence risk in patients with solitary HCC ≤ 5 cm who underwent SR or TA. Additionally, we explored the potential role of these models in personalized therapeutic decision-making to promote the advancement of precision medicine in HCC management.

Materials and methods

Patients

This study was conducted in accordance with the Declaration of Helsinki and approved by two institutional review boards (No. B2022-223R and No. 2022-140-01). Given the retrospective observational nature of the study without involving any direct patient contact or intervention, the informed consent was waived with the permission of the ethics committee. Patient records/information were anonymized and de-identified prior to analysis. From January 2017 to October 2022, HCC patients in two medical centers who met the following criteria were consecutively enrolled. The inclusion criteria were as follows: (i) HCC (≤ 5 cm, solitary, without macrovascular invasion or extrahepatic metastasis) patients with pathologically or clinically confirmed; (ii) no previous history of

treatment or puncture, and the initial treatment was SR or TA; (iii) CEUS was performed within 2 wk prior to SR or TA. The exclusion criteria were as follows: (i) incomplete SR or TA; (ii) any postoperative adjuvant therapy such as TACE or targeting or immunotherapy; (iii) the follow-up time was less than 1 y; (iv) poor quality of CEUS images or incomplete clinical data; (v) other malignant tumors except HCC. Finally, a total of 575 patients were enrolled in this study, including 365 cases in the SR group and 210 cases in the TA group. The patient enrollment flow chart is shown in Figure 1.

Demographic and clinical characteristics

Baseline patient characteristics during the 2 wk prior to SR or TA were collected from electronic medical records. The collected parameters included: (i) demographic variables (sex, age); (ii) tumor characteristics (location, size); (iii) hepatic comorbidities (cirrhosis, fatty liver); (iv) viral serology (hepatitis B surface antigen [HBsAg], anti-hepatitis C virus antibody [anti-HCV]), (v) liver and renal function assessments (albumin-bilirubin [ALBI] grade, total bilirubin [Tbil], albumin [ALB], alanine aminotransferase [ALT], aspartate aminotransferase [AST], alkaline phosphatase [ALP], gamma-glutamyl transferase [GGT], serum creatinine [Scr]; prothrombin time [PT], international normalized ratio [INR]), (vi) blood routine and coagulation function (hemoglobin [Hb] and platelet count [PLT], prothrombin time [PT], international normalized ratio [INR]), and (vii) tumor biomarkers (alpha-fetoprotein [AFP], abnormal prothrombin [APT]). ALBI scores were calculated using the previously established formula [15]: $ALBI = (\log_{10} Tbil [\mu\text{mol/L}] \times 0.66) + (ALB [g/L] \times -0.085)$. Patients were divided into three ALBI grades: Grade 1 (ALBI score ≤ -2.60), Grade 2 ($-2.60 < \text{ALBI score} \leq -1.39$), and Grade 3 (ALBI score > -1.39).

HCC treatment and patient follow-up

Treatment decisions were made by a multidisciplinary team based on disease status and patient preferences. All the patients who underwent SR were confirmed by pathology to have achieved R0 resection, meaning that there were no tumor cells remaining at the surgical margins under the microscope. After SR or TA, all patients were followed up according to outpatient review, including abdomen-pelvis ultrasound, contrast-enhanced computed tomography (CECT) or contrast-enhanced magnetic resonance imaging (CEMRI), as well as laboratory tests (including liver function and serum AFP level). Follow-up was conducted 1 mo after the initial treatment, every 3 mo during the first 2 y and every 6 mo thereafter. The endpoint of the study was recurrence-free survival (RFS), defined as the time interval from the date of SR or TA to the date of first recurrence or last visit. The last follow-up date was October 2024. Recurrence was determined by at least two different imaging features or histopathological examination.

Image acquisition and preprocessing

All the ultrasound examinations were performed by radiologists with more than 5 y of experience in CEUS and were conducted in accordance with standardized operational protocols. Different ultrasonic instruments were used in this study, including LOQIQ E9, Philips EPIQ7 and Aplio i900. B-mode ultrasound (BMUS) scans were performed first, and then switched to contrast mode. A bolus of 1.–2.4mL SonoVue (Bracco Imaging, Milan, Italy) was injected into the antecubital vein, followed by 5 mL saline flush. The images and videos, including the arterial phase (AP, 10–45 s), portal venous phase (PVP, 45–120 s) and delayed phase (DP, >120 s), were stored in DICOM format for subsequent analysis. One frame was selected from each of BMUS, AP, PVP and DP. The image selection criteria were as follows: In the case of BMUS, the image with the highest clarity was selected. For CEUS, the AP image was chosen at the time corresponding to the peak enhancement. The images of PVP and DP were selected 90 s and 150 after the injection of the contrast

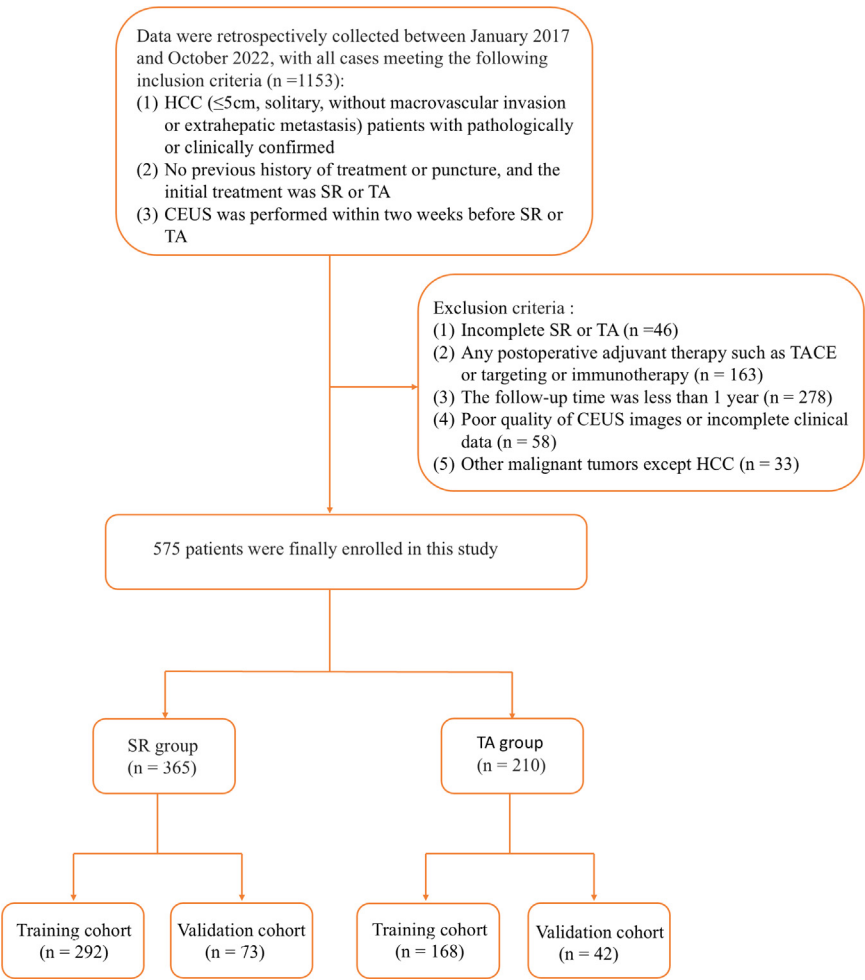


Figure 1. The flowchart of patient inclusion and exclusion.

agent, respectively. The tumor region of interest (ROI) was manually drawn along the tumor margin by a professional ultrasound radiologist (K.K.C.) with 5 y of experience using 3D Slicer software (version 5.7.0, <https://www.slicer.org/>). Additionally, the peri-tumoral liver tissue was automatically segmented by expanding 5 mm outward from the tumor margin. Given the limited sample size, data augmentation techniques were used to preprocess the images during model training to mitigate potential overfitting of the network.

Model development

The dataset were partitioned into the SR group and the TA group based on the treatment modalities. Each group was randomly divided into the training and the validation cohorts in a ratio of 8:2. We designed a dual-stream visual encoder based on a ViT architecture initialized with CLIP image encoder pretrained weights to extract multi-modal image features. These image features were then fused with clinical features, leveraging their complementarity among these modalities to develop prognostic prediction models (ViT-SR for the SR group and ViT-TA for the TA group). Each modality was encoded independently through a dedicated encoder with shared backbone weights. The model was trained using the ADAMW optimizer with weight decay ($\lambda = 0.0001$). The initial learning rate for ViT-SR was set to 5.0×10^{-5} , while for ViT-TA it was 3.0×10^{-3} . Step decay scheduler ($\gamma = 0.1$) was configured. The learning rate of ViT-SR started to decay from epoch 25, and that of ViT-TA started from epoch 30. The batch size for training was set to 256, and the total training duration was 30 epochs. This method initially segmented images from different phases into fixed-sized image patches, which served as the input units for the Transformer network. Supplemental material described the visual feature extraction method in detail.

Concurrently, the patient’s clinical data were transformed into high-dimensional vector representations via an embedding module. These representations were further processed by a multilayer perceptron for feature selection and dimensionality reduction before being fused with the image features. The fused features were passed through three fully connected layers, followed by a Sigmoid activation function to generate the patient’s risk prediction score. Based on the prediction results of the Cox regression model, the training and validation cohorts were stratified into high-risk and low-risk groups according to the median risk score. To assess the contribution of each input modality, we conducted a series of ablation experiments by training the ViT-SR and ViT-TA models using only BMUS, only CEUS, or only clinical variables, respectively. The combined model integrating all modalities (BMUS + CEUS + clinical) served as the reference. All training configurations, including data augmentation, optimizer settings, and training epochs, were kept identical to ensure fair comparison across modalities. The flowchart of ViT analysis is shown in Figure 2.

Model visualization

To improve the interpretability of the model, we employed Gradient-Weighted class activation mapping (Grad-CAM) to further visualize critical regions of the model. Heatmaps were generated by computing gradient-weighted averages of self-attention features from the final ViT layer.

Exploratory assessment of treatment conversion benefits

To minimize selection bias between different treatment groups and make the results of the simulated treatment switching more representative of the true discriminative ability of the model, we employed a 1:1

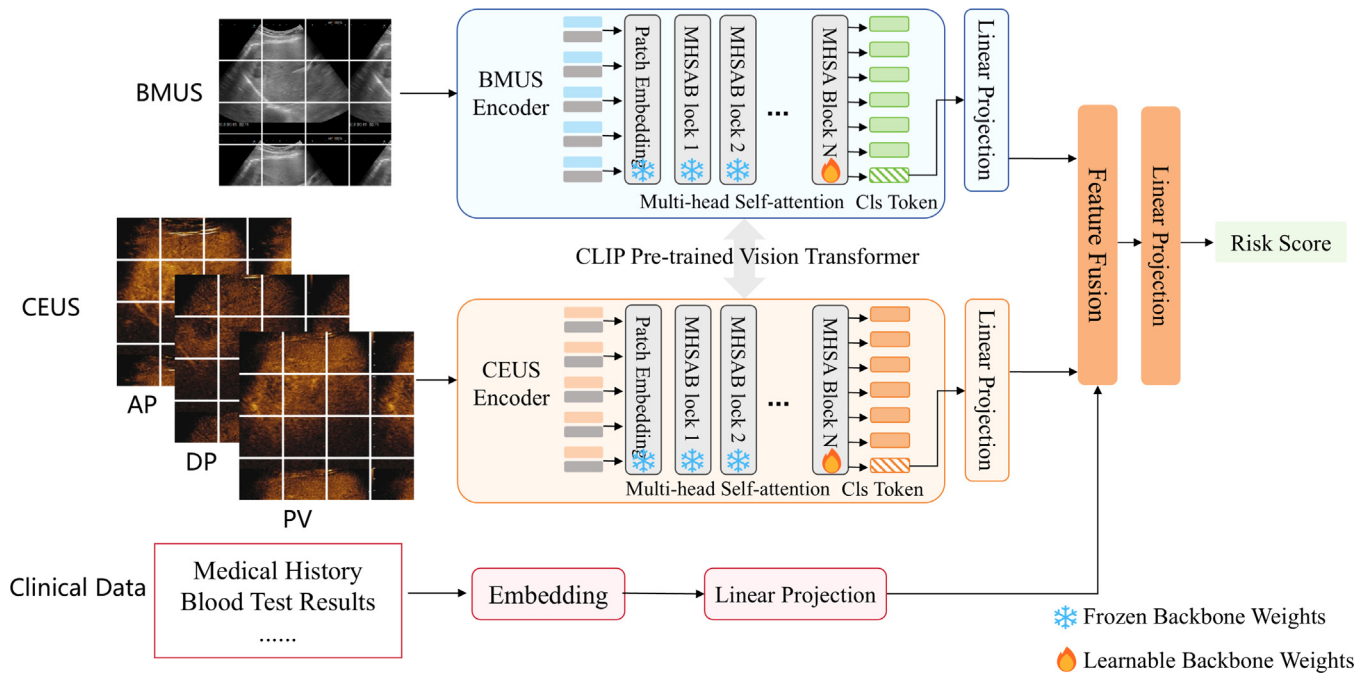


Figure 2. The flowchart of ViT analysis. AP, arterial phase; DP, delayed phase; PV, portal venous phase; ViT, Vision Transformer.

propensity score matching (PSM) with a caliper of 0.2 to construct balanced sub-cohorts of SR and TA with comparable baseline characteristics. Patients from each treatment cohort were incorporated into the opposing ViT model to simulate their risk profiles under the alternative treatment, resulting in new risk stratifications. We focused on patients who may have derived potential benefit from this process, specifically those reclassified from high to low recurrence risk.

Statistical analysis

All statistical analyses were performed using R software (version 4.3.0, <http://www.r-project.org/>) and Python (version 3.7, <https://www.python.org/>). Continuous variables following a normal distribution were expressed as mean $\pm$ standard deviation (SD) and compared using Student's *t*-test. Otherwise, they were displayed as median (interquartile range) and compared using Mann–Whitney U test. Categorical variables were presented as frequency (percentage) and were compared by χ^2 test or Fisher's exact test. Model performance was assessed using the C-Index, area under the receiver operating characteristic (ROC) curve (AUC), precision and F1 score. The dynamic predictive performance of the model was evaluated using the time-dependent ROC curves and calibration curves. The RFS rates between subgroups were visualized by Kaplan–Meier curve and differences were compared using the log-rank test. Two-sided *p* < 0.05 was considered statistically significant.

Results

Patient characteristics

Patient characteristics are shown in Table 1. The median follow-up period was 33.6 mo in the SR group and 23.4 mo in the TA group. During the follow-up period, a total of 115 patients (115/365, 31.5%) in the SR group experienced recurrence, including local tumor progression (28/115, 24.4%), intrahepatic metastasis (81/115, 70.4%), and extrahepatic metastasis (6/115, 5.2%). In contrast, a total of 112 patients (112/210, 53.3%) in the TA group experienced recurrence, including local tumor progression (61/112, 54.5%), intrahepatic metastasis (50/112, 44.6%) and extrahepatic metastasis (1/112, 0.9%). The 1-, 3-, and 5-y RFS rates in the SR group were 90%, 69% and 58%, respectively,

whereas the corresponding rates in the TA group were 79%, 50%, and 37%, respectively. Patient characteristics between the training and validation cohorts are shown in Table S1.

Model performance

The detailed predictive performances were provided in Table 2. The fivefold cross-validation results of the combined models are shown in Table 3. Across both SR and TA cohorts, single-modality models showed

Table 1

Preoperative clinical characteristics of patients in the TA and SR groups.

Characteristics	SR (n = 365)	TA (n = 210)	p Value
Sex (male/female)	305/60	153/57	0.002
Age (y)	57.61 $\pm$ 10.89	60.55 $\pm$ 11.60	0.002
Location ^a	83/266/10/5	41/155/12/2	0.335
Tumor size (mm)	25.00 (16.00–36.00)	18.00 (13.00–22.00)	< 0.001
Cirrhosis ^b	150/215	43/167	< 0.001
Fatty liver ^b	299/66	187/23	0.023
HBsAg ^c	85/280	58/152	0.247
Anti-HCV ^c	354/11	200/10	0.282
ALBI grade (1/2/3)	318/47/0	145/64/1	< 0.001
Tbil (μ mol/L)	13.40 (10.30–17.35)	14.20 (10.85–19.60)	0.033
ALB (g/L)	43.00 (41.00–46.00)	43.00 (38.00–46.00)	0.005
ALT (U/L)	25.00 (19.00–35.50)	31.00 (20.00–44.00)	0.005
AST (U/L)	24.00 (19.00–33.00)	33.00 (24.75–41.25)	< 0.001
ALP (U/L)	73.00 (62.00–90.00)	89.00 (69.75–108.25)	< 0.001
GGT (U/L)	33.00 (20.50–57.00)	42.50 (26.00–79.25)	< 0.001
Scr (μ mol/L)	77.00 (66.00–85.50)	71.00 (60.75–82.25)	0.001
APT (mAU/mL)	63.00 (28.00–298.50)	52.00 (25.75–108.00)	< 0.001
AFP (ng/mL)	11.40 (3.10–148.00)	18.20 (4.70–103.45)	0.156
PT (s)	11.90 (11.50–12.50)	12.40 (11.70–13.40)	< 0.001
INR	1.06 (1.02–1.10)	1.09 (1.03–1.16)	< 0.001
Hb (g/L)	143.00 (135.00–153.00)	138.00 (125.75–150.00)	< 0.001
PLT ($\times 10^9$ /L)	154.00 (115.00–194.00)	117.00 (73.00–164.25)	< 0.001

^a Left/right/junction of left and right/caudate lobe;

^b Absent/present;

^c Negative/positive. Categorical variables were expressed in counts, and continuous variables were expressed in means $\pm$ SD or medians (interquartile ranges). Bold values are variables with *p* < 0.05.

Table 2
The predictive performance of the models in validation cohorts

Groups	Modal	C-Index	AUC	Precision	F1-Score
ViT-SR	BMUS	0.63	0.63	0.66	0.58
	CEUS	0.60	0.61	0.64	0.55
	Clinical	0.61	0.60	0.68	0.56
	Combined	0.73	0.76	0.86	0.70
ViT-TA	BMUS	0.62	0.66	0.71	0.63
	CEUS	0.63	0.67	0.72	0.64
	Clinical	0.66	0.53	0.64	0.50
	Combined	0.76	0.83	0.76	0.76

Bold values indicate the performance of the combined models in the validation cohorts.

consistently lower performance than the combined model. The C-Index was 0.73 for the combined ViT-SR and 0.76 for the combined ViT-TA group. The RFS of the low-risk subgroups was significantly prolonged in both the SR and the TA group across both the training and validation cohorts (Fig. 3, $p < 0.05$ for all cohorts). These findings highlighted the model’s robust stratification capability. In both training and validation sets, the 12-mo recurrence-free probability exceeded 90% in the TA low-risk subgroup, comparable to SR group outcomes, demonstrating the efficacy of TA for this subgroup and supporting its role as an appropriate alternative when low-risk patients opt against SR treatment. However, for high-risk patients, SR consistently achieved significantly superior 12-

Table 3
The fivefold cross-validation results of the combined models in validation cohorts

Groups	Fold	C-Index	AUC	Precision	F1-Score
ViT-SR	1	0.73	0.76	0.86	0.70
	2	0.71	0.74	0.84	0.67
	3	0.75	0.78	0.87	0.72
	4	0.72	0.75	0.83	0.68
	5	0.74	0.77	0.88	0.73
	Mean	0.73 ± 0.02	0.76 ± 0.02	0.86 ± 0.02	0.70 ± 0.03
ViT-TA	1	0.76	0.83	0.76	0.76
	2	0.74	0.81	0.72	0.73
	3	0.78	0.85	0.79	0.78
	4	0.77	0.84	0.77	0.77
	5	0.75	0.82	0.74	0.74
	Mean	0.76 ± 0.02	0.83 ± 0.02	0.76 ± 0.03	0.76 ± 0.02

mo recurrence-free probability versus TA across both datasets. Figure 4 revealed that the time-dependent AUC values showed favorable stability across 1-, 3-, and 5-y follow-up periods (ViT-SR: 0.76, 0.77, 0.76; ViT-TA: 0.79, 0.84, 0.82). The calibration curves for both models demonstrated good overall agreement between the predicted and observed probabilities of 1-, 3-, and 5-y RFS in the validation cohort (Figure S1). Minor deviations were observed in a few low-probability bins,

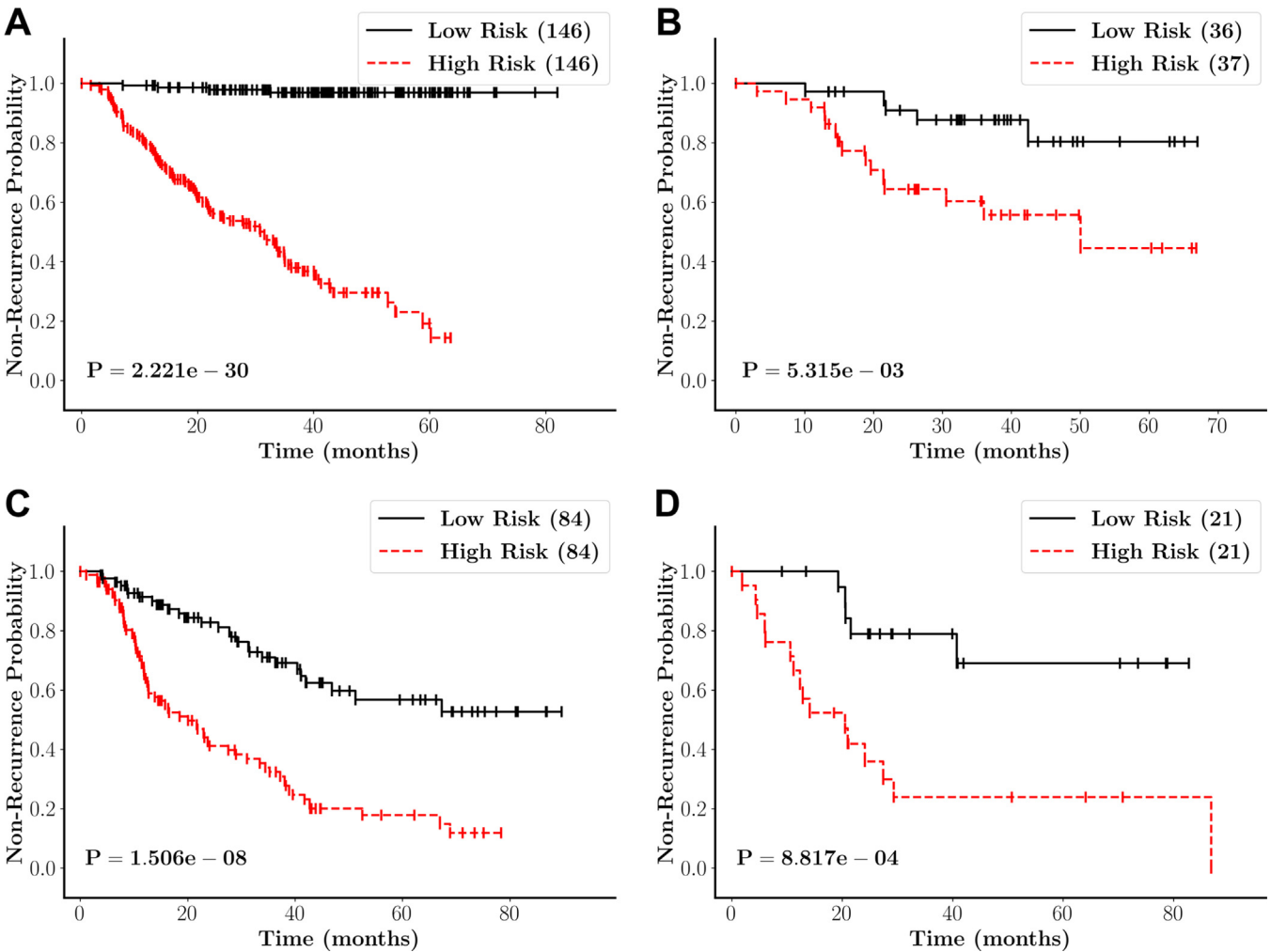


Figure 3. Kaplan–Meier survival curves between the low- and high-risk subgroups. Kaplan–Meier survival curves of combined ViT-SR in the training (a) and validation cohorts (b). Kaplan–Meier survival curves of combined ViT-TA in the training (c) and validation cohorts (d). SR, surgical resection; TA, thermal ablation; ViT, Vision Transformer.

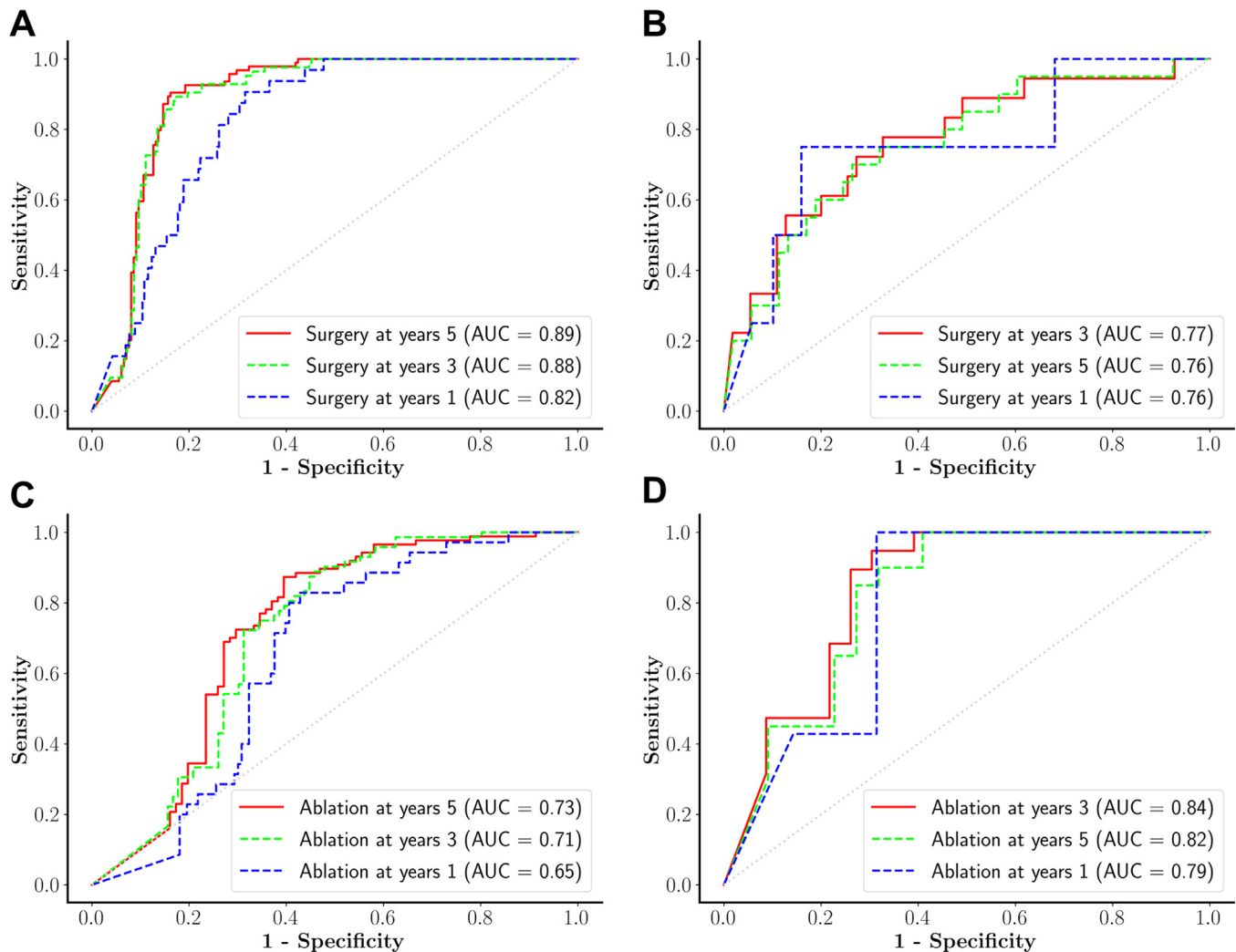


Figure 4. Time-dependent ROC curve analyses for the predictive efficacy of RFS at distinct follow-up intervals (1-, 3-, and 5-y) in the SR group (a for training cohort; b for validation cohort) and the TA group (c for training cohort; d for validation cohort). RFS, recurrence-free survival; ROC, receiver operating characteristic; SR, surgical resection; TA, thermal ablation.

particularly for the 5-y horizon, likely due to fewer long-term events and increased censoring.

Model explainability

Figure 5 depicts the suspicious areas found by the ViT model. These highlighted regions, particularly those visualized in red or yellow tones, were closely associated with recurrence risks. Additionally, the green and blue areas signified a gradual reduction in predictive relevance. In both SR and TA cohorts, high-risk cases showed model attention concentrated on irregular tumor margins, close association with blood vessels and heterogeneous internal echotexture on BMUS. In CEUS, the model predominantly focused on heterogeneous AP hyperenhancement and early washout in the PVP. Conversely, low-risk cases consistently showed attention on well-defined margins, homogeneous AP hyperenhancement, and delayed or mild washout.

Subgroup analysis by TA type

The TA cohort was divided into subgroups based on ablation technique: 147 patients underwent radiofrequency ablation (RFA), and 63 underwent MWA. Subsequently, we compared the recurrence risk scores predicted by the ViT-TA model between the different subgroups. As shown in the violin–box plots (Figure S2), the predicted recurrence risk

scores did not differ significantly between patients receiving RFA and MWA ($p = 0.957$).

Exploratory assessment of treatment conversion benefits

After PSM, both the SR and TA subgroups included 146 patients each, with their baseline characteristics presented in Table S2. And the model maintained good risk stratification performance in the matched cohort (Figure S3). We subsequently evaluated the possibility of identifying patients who might benefit from hypothetical treatment switching in the cohorts after PSM. We found that under the hypothetical scenario of switching from SR to TA, 21.2% (31/146) low-risk and 38.4% (56/146) high-risk patients remained in their original risk subgroups. However, 40.4% patients (59/146) experienced risk category reclassification, with 11.6% (17/146) being downgraded from high- to low-risk, which indicated TA-mediated possible risk attenuation. Conversely, transitioning from TA to SR resulted in 41.4% (60/146) of low-risk patients being upgraded to high-risk and 4.8% (7/146) of high-risk patients being downgraded to low-risk (Fig. 6). We analyzed the clinical and serological characteristics of patients who were expected to benefit from the change in treatment strategy, and the results are presented in Table S3.

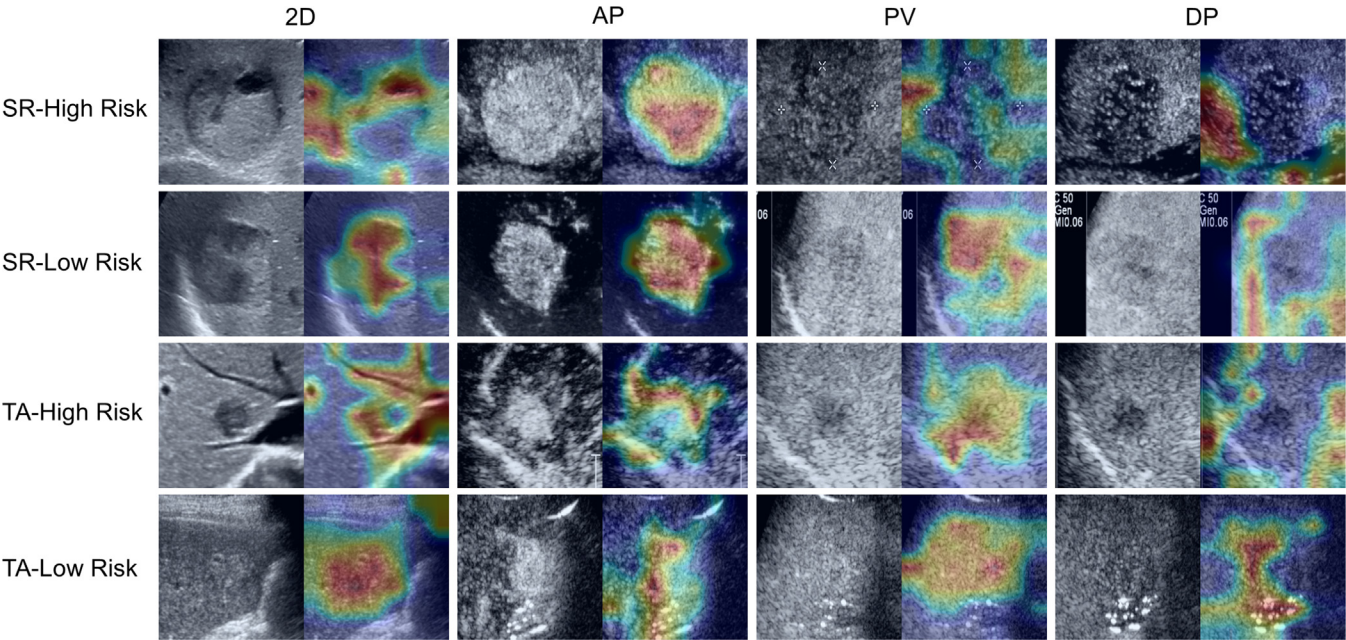


Figure 5. The input ROI images with a heatmap calculated by the Grad-CAM method. ROI, region of interest.

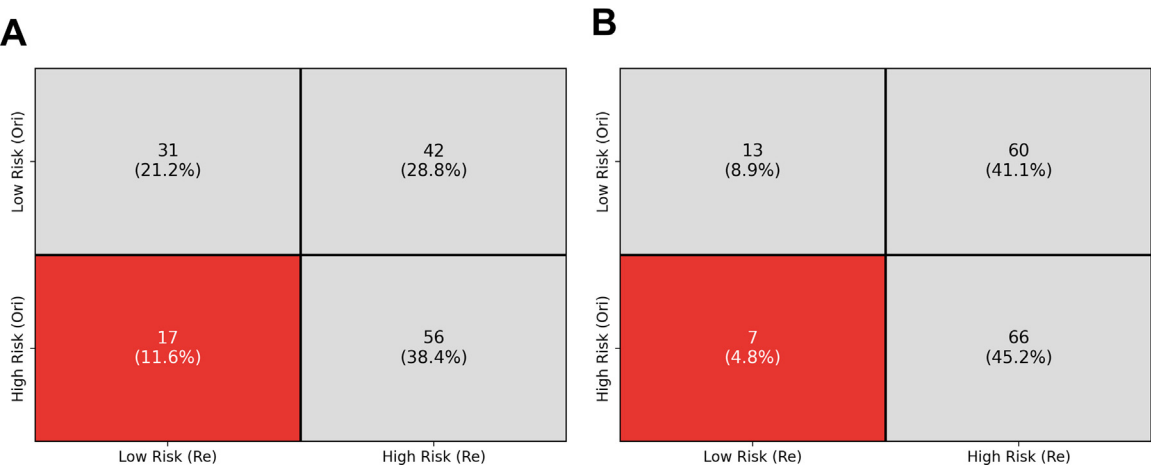


Figure 6. Risk reclassification by simulating treatment strategy exchanges. (a) Original stratification (Ori) results from combined ViT-SR, and re-stratification (Re) results from combined ViT-TA. Seventeen patients (red part) were stratified as being at high risk for receiving SR, yet at low risk when switching to TA. (B) Original stratification (Ori) results from combined ViT-TA and re-stratification (Re) results from combined ViT-SR. Seven patients (red part) were stratified as being at high risk for receiving TA and at low risk when switched to SR. SR, surgical resection; TA, thermal ablation; ViT, Vision Transformer.

Discussion

In this study, we developed ViT-based models integrating multi-modal ultrasound images and clinical data to predict recurrence risk in patients with solitary HCC ≤ 5 cm receiving either SR or TA, while also serving as an exploratory tool to aid clinical discussions. The ablation experiments highlighted the critical importance of incorporating multi-source input information. BMUS and CEUS capture different but complementary aspects of tumor biology—structural morphology and vascular perfusion—while clinical variables provide additional context related to liver function and tumor aggressiveness. Individually, each modality contributed only partial prognostic information, resulting in inferior predictive performance. The full multi-modal architecture, however, achieved significantly improved performance, suggesting that integrating heterogeneous sources enhances the model’s ability to characterize recurrence-associated imaging phenotypes.

Our combined models demonstrated favorable predictive performance, with C-indices of 0.73 (ViT-SR) and 0.76 (ViT-TA), indicating strong concordance between predicted and observed RFS. The AUC values of the combined models for ViT-SR and the ViT-TA were 0.76 and 0.83, respectively, showing the excellent risk stratification ability. The time-dependent AUC values for 1–5-y RFS prediction exhibited fluctuations with <0.05 , confirming the consistent and reliable prognostic ability across short-, medium-, and long-term follow-up periods of the models. This robustness likely stems from the unique self-attention mechanism of ViT, which enables the capture of long-range dependencies and global feature interactions, thereby achieving superior multi-modal fusion efficacy. Notably, the combined ViT-SR demonstrated minimal fluctuation in AUC values, likely attributable to the definitive nature of SR, which mitigated heterogeneous recurrence factors. Additionally, we found that combined ViT-TA had a slightly lower 1-y AUC than ViT-SR but increased significantly over time, demonstrating better long-term performance. Furthermore, the analysis of the heterogeneity

of ablation techniques revealed that the risk scores predicted by the ViT-A model showed no significant differences among different ablation methods.

HCC is a highly heterogeneous malignancy that interacts dynamically with the tumor microenvironment, leading to prognostic divergence [9,16]. Its high post-treatment recurrence risk remains a worldwide challenge [17]. Accumulating evidence substantiates the value of CEUS features in predicting prognosis after TA or SR [18–22]. However, conventional imaging analysis remains constrained by its reliance on subjective visual interpretation and operator-dependent assessments, which may lead to overlooking or insufficient quantification of some key details. Quantifying image information through radiomics significantly minimizes subjective bias in image analysis [23]. Currently, there are not many studies applying DL to CEUS images for predicting the prognosis of HCC. Zhang et al developed a ResNet-50-based model using CEUS images from 172 HCC patients who underwent SR to predict early recurrence (ER), achieving an AUC of 0.834 [24]. Huang et al. [25] constructed a DL model based on grayscale and CEUS images from 414 HCC patients after SR, which predicted ER with an AUC of 0.759. Zhang et al built a ViT-based DL model using CEUS and MRI images from 401 HCC patients treated with multiple TA, reporting an AUC of 0.87 in the internal validation cohort for predicting ER [26]. To date, there have been no studies investigating the value of CEUS and ViT-based DL in optimizing HCC treatment strategies, and our study fills this gap.

Methodologically, the ViT framework's global attention mechanism outperforms traditional CNNs in capturing multi-modal information. Its global self-attention mechanism enables more effective integration of multi-phase CEUS features and clinical variables, capturing long-range spatial dependencies that conventional CNN-based or radiomics approaches cannot. This leads to more stable risk stratification across different follow-up periods and improves the detection of subtle peritumoral perfusion patterns associated with recurrence [12]. From a clinical application perspective, the self-attention mechanism of ViT distinctly highlights the contribution of local regions to the model's predictions. When visualized by Grad-CAM, this approach significantly improves model interpretability—addressing a critical limitation of “black-box” AI systems in clinical oncology. This advancement enables clinicians to visualize region-specific contributions, thereby providing more intuitive decision-making support. However, quantitative evaluation of ViT attention maps is not feasible due to the lack of pixel-level prognostic ground truth and the absence of validated metrics for attention calibration. Therefore, following standard practice, we provide qualitative radiologist-reviewed attention visualizations to illustrate that the model focuses on clinically relevant regions. The imaging features highlighted by our model, including tumor margins, vascular relationships, echo heterogeneity, heterogeneous AP hyperenhancement and early washout, showed partial alignment with clinically recognized high-risk features associated with recurrence. Regular shapes typically indicate expansive growth of the tumor. While irregular shapes often suggest invasive growth patterns, which make the tumor more likely to break through the pseudo-capsule and invade the surrounding liver parenchyma. Existing studies have shown that irregular tumor shapes are associated with microvascular invasion (MVI), and patients with such tumors have lower OS and RFS rates than HCC patients with regular tumor shapes [27]. Close tumor-vessel association may also indicate vascular invasion risk. And, this might cause the thermal sink effect during TA, thereby resulting in a poorer effect. Tumors show internal echo heterogeneity on BMUS and heterogeneous hyperenhancement patterns during the AP on CEUS, indicating internal heterogeneity or necrosis. He et al. demonstrated that heterogeneous enhancement within lesions during the AP correlated with higher tumor proliferative activity and shorter recurrence time [28]. Furthermore, the model's focus on early washout aligns with findings by Wang et al. They noted that the “rapid washout” enhancement pattern was an independent risk factor for early postoperative recurrence, and the washout time was related to tumor

differentiation, with shorter washout times indicating poorer differentiation [29]. In conclusion, the ViT model appeared capable of learning prognostically significant features from ultrasound images, which contributed to improved interpretability of the model's predictions.

This study systematically assessed the postoperative recurrence probability of SR and TA in HCC patients across various risk stratifications using Kaplan–Meier survival curve analysis. We found that high-risk patients exhibited a sharp decline in survival curves within 12 mo after TA, indicating limitations of TA treatment in controlling ER for this subgroup, while SR would be a preferable therapeutic option. Furthermore, in the risk reclassification analysis of the baseline-balanced cohort, we observed that some patients might obtain clinical benefits from shifting treatment approaches. By summarizing the clinical characteristics of such patients, we found that those transitioning from high-risk in TA to low-risk in SR tended to be younger (mean age 48 y) and exhibited higher AFP and APT levels. However, patients moving from the high-risk in TA to the low-risk in SR often demonstrated opposite outcomes. A retrospective study by Zhang et al. [30] on outcomes following curative liver resection in adolescent and young (AYA) patients with HCC demonstrated that compared to the non-AYA group, the AYA group exhibited higher AFP levels, more unfavorable pathological features, and poorer disease-free and OS. Furthermore, elevated AFP and APT levels have been consistently associated with recurrence risk across multiple studies [31,32]. Therefore, these indicators collectively suggest that patients who may benefit from shifting from TA to SR strategies likely have a relatively higher recurrence risk. This aligns with our general clinical understanding, as patients at high risk of recurrence, when opting for SR treatment, can achieve better tumor clearance and removal of potential satellite lesions due to wider margins compared to TA. Similarly, Bai et al. [33] reported that liver resection, compared to percutaneous RFA, significantly reduces early and local recurrence risks in high-risk HCC patients stratified by microvascular invasion. However, this potential benefit still requires validation in future prospective randomized controlled trials.

Despite the encouraging findings, several limitations warrant further exploration and improvement. First, as this was a retrospective study, the non-random assignment of treatments introduced selection bias. Although we employed active data augmentation, regularization techniques (Dropout and weight decay), and early stopping during training to mitigate overfitting and potential allocation bias to some extent, future prospective multicenter collaborations are needed to collect larger sample sizes and further reduce selection bias. Second, in real-world clinical practice, each patient receives only one treatment, meaning there are no actual outcomes available for the alternative therapy. Therefore, the risk reclassification analysis was based on observational data rather than counterfactual inference. Consequently, this analysis represented a correlation-based prediction and cannot be directly equated with “treatment benefit.” Our study served as an exploratory effort that may provide a basis and inspiration for future research. Prospective randomized controlled trials or dedicated causal frameworks are warranted in future studies to further achieve counterfactual validation. Third, although CEUS is inherently a dynamic imaging modality, our study extracted only a single representative frame from each phase. This approach, while pragmatic for standardization across a retrospective cohort, does not fully leverage the rich temporal information contained in full CEUS video sequences. Future studies with prospective, standardized video acquisition protocols may explore the additional prognostic value of incorporating full temporal dynamics using advanced video-based DL architectures. Fourth, due to the lack of long-term survival outcome data for patients, this study did not include OS as an indicator. However, relying solely on RFS may not fully capture all dimensions of clinical benefit. Fifth, larger prospective datasets are still needed to further verify the potential interaction between ablation modalities and risk scores. Finally, it is worth mentioning that multi-modal imaging combining ultrasound with MRI or CT may have the potential to improve model performance, warranting further investigation.

Conclusions

In conclusion, we developed two ViT-based models to predict the recurrence risk of solitary HCC ≤ 5 cm following SR or TA. By simulating treatment conversion, this approach may facilitate discussions on personalized clinical treatment strategies.

Data availability statement

The datasets generated or analyzed during the study are available from the corresponding author on reasonable request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 82472004 and No. 82272013), Clinical Trials from the Affiliated Drum Tower Hospital, Medical School of Nanjing University (2022-LCYJ-MS-24) and Departmental Research Fund from Affiliated People's Hospital of Jiangsu University (KFB2025007). All funding departments had no role in the collection, analysis, or interpretation of the data or in the decision to submit the manuscript for publication.

Declaration of competing interest

The authors declare no competing interests.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ultrasmedbio.2026.03.028.

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